

$$1) \det(A) = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{vmatrix} = 1(2-0) - 2(0+1) = 2-2=0$$

$$= 1(2-0) + 1(-2-0) = 2-2=0$$

$$= 1(-2-0) + 2(1-0) = -2+2=0$$

a) $\det(2A) = 2^3 |A| = 8(0) = 0$

b) $\det(A^{-1}) = \frac{1}{\det(A)}$

c) $\det(A^t) = \det(A) = 0$

d) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -2 & 2 \end{pmatrix} \xrightarrow{\begin{matrix} R_3 \rightarrow R_3 + 2R_2 \\ R_2 \rightarrow R_2 - 2R_1 \end{matrix}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

e) $\text{Rank}(A) = 2$

f) $Ax = \lambda x \Rightarrow (A - \lambda I)x = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 1-\lambda & -1 \\ 1 & 0 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-3\lambda+\lambda^2) - 2(1) =$

$$= 1 - 3\lambda + \lambda^2 - 2 + 6\lambda - 2\lambda^2 = -1 + 3\lambda - \lambda^2 = -(\lambda^2 - 3\lambda + 1) = -(\lambda - \frac{3}{2})^2 + \frac{5}{4}$$

$$= 0 \Rightarrow \lambda = 0 \text{ or } \lambda = \frac{3 \pm \sqrt{5}}{2} \Rightarrow \lambda = 0$$

2) 'B' is singular iff $|B| \neq 0$. True

3) $\lim_{x \rightarrow 5} \left(\frac{\sqrt{x} - \sqrt{5}}{x^2 - 25} \right) = \lim_{x \rightarrow 5} \left(\frac{\sqrt{x} - \sqrt{5}}{(x-5)(x+5)(\sqrt{x} + \sqrt{5})} \right) = \lim_{x \rightarrow 5} \left(\frac{1}{(x+5)(\sqrt{x} + \sqrt{5})} \right) = \frac{1}{10(2\sqrt{5})} = \frac{1}{20\sqrt{5}} = \frac{\sqrt{5}}{100}$

4) $f(x) = \begin{cases} mx+3 & \text{if } x > 1 \\ 4 & \text{if } x = 1 \\ x^2+n & \text{if } x < 1 \end{cases}$. $m=?$, $n=?$ If f is continuous?

$$\lim_{x \rightarrow 1^+} (f(x)) = m(1) + 3 = m + 3 = f(1) = 4 \Rightarrow m = 1$$

$$\lim_{x \rightarrow 1^-} (f(x)) = 1^2 + n = 1 + n = f(1) = 4 \Rightarrow n = 3$$

Part II

1) $\begin{cases} x + ay = 4 \\ ax + 9y = 6 \end{cases}$ Gaussian elimination.

a) Unique

b) Infinitely many

c) No

2) $\lim_{x \rightarrow 3} (x^2 + 1) = 10$ $\epsilon - \delta$

3) $p(x) = 2x^3 - 3x^2 + 4$ has a root in $(-1, 1)$. Show using intermediate value theorem.